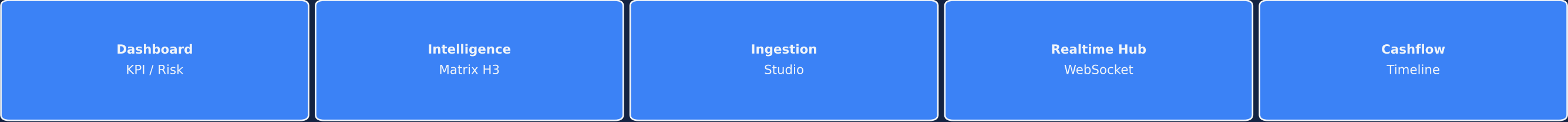
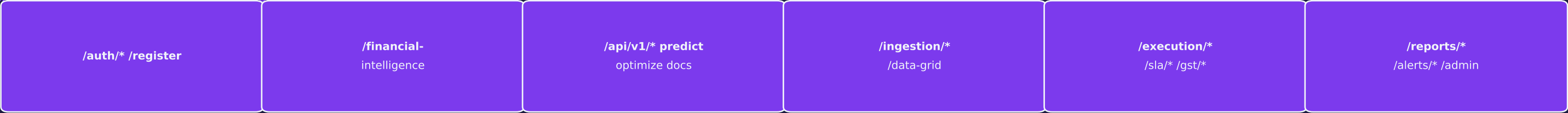


1 · High-Level System Architecture

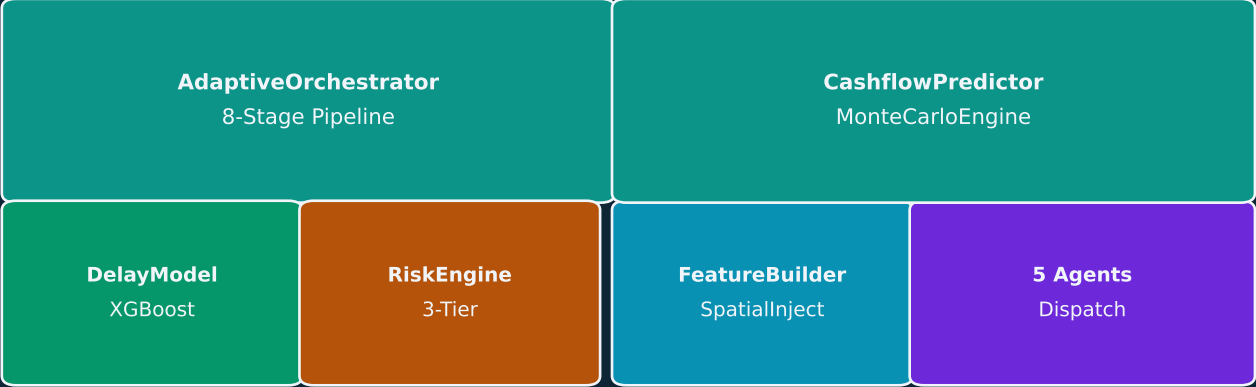
CLIENT LAYER (React + Vite · axios · React Query)



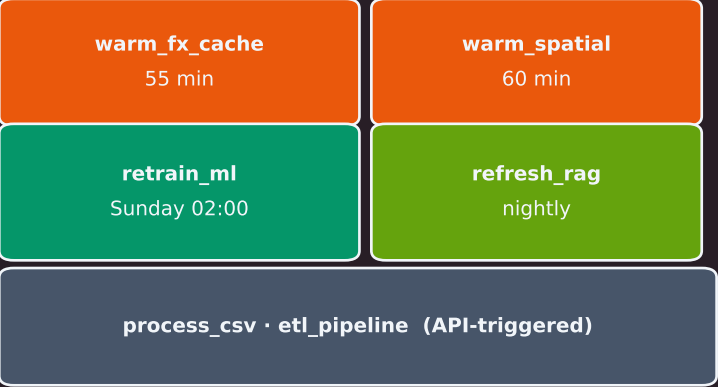
API GATEWAY LAYER (FastAPI · uvicorn · SlowAPI rate limiting · CORS middleware)



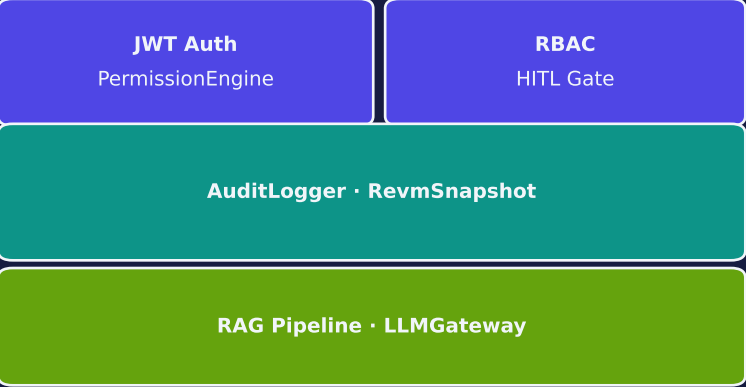
FINANCIAL INTELLIGENCE ENGINE



ASYNC TASKS (Celery + Redis)



AUTH & GOVERNANCE



DATA LAYER



EXTERNAL INTEGRATIONS





3 · Risk Scoring — 3-Tier Inference Architecture

INPUT · {route, carrier, order_value, total_cost, credit_days, cargo_type, industry_vertical, delay_days}

STEP 0 · Spatial Enrichment (spatial_risk_injector.py)

SpatialRiskInjector

.get_route_severity(route,
cargo_type)

Redis cache (TTL 5 min)

→ spatial_severity: float [0.0 - 1.0]

_route_to_h3_cells('CN-EU_SUEZ') tokens: [CN, EU, SUEZ]
CN→H3(31.23,121.47,r3) EU→H3(51.90,4.48,r3) SUEZ→H3(29.97,32.55,r3)
for each event: contribution = severity × 0.55^h3_distance (MAX_HOPS=3)
max_severity = max(all contributions across all route cells)

STEP 1 · Feature Engineering (feature_engine.py)

Derived Scalars

cost_ratio = total_cost / order_value
margin_pct = margin / order_value
carrier_reliability (CARRIER_RELIABILITY_REGISTRY)
route_complexity (1.0-4.0) · holding_rate (cargo type)
seasonality_idx (industry_vertical × month)

Interaction Terms (9 features)

cost_x_delay · carrier_x_route · spatial_x_route · spatial_x_cargo
margin_x_spatial · seasonal_x_cargo · credit_delay_exposure
margin_compression = margin_pct / max(cost_ratio, 0.01)
risk_pressure = cost_ratio × carrier_risk × route_complexity ← highest-signal 3-way term

TIER 1 · XGBoost (xgb_model.pkl)

XGBClassifier

n_est=150 · max_depth=4
scale_pos_weight=auto
eval_metric=logloss

ColumnTransformer

(passthrough + OHE route/carrier)

spatial_lift = severity × 0.25 × (1 - base_prob)
final = base_prob + lift

pkl missing

TIER 2 · Gradient Boosting (gb_model.pkl)

GradientBoostingClassifier

n_est=120 · max_depth=4
subsample=0.8
min_samples_leaf=20

risk_fallback.pkl

23-col numeric input (no OHE)

Loads: risk_fallback_columns.pkl
Aligns feature order at runtime

fallback missing

TIER 3 · Pure Math

sigmoid(logit)

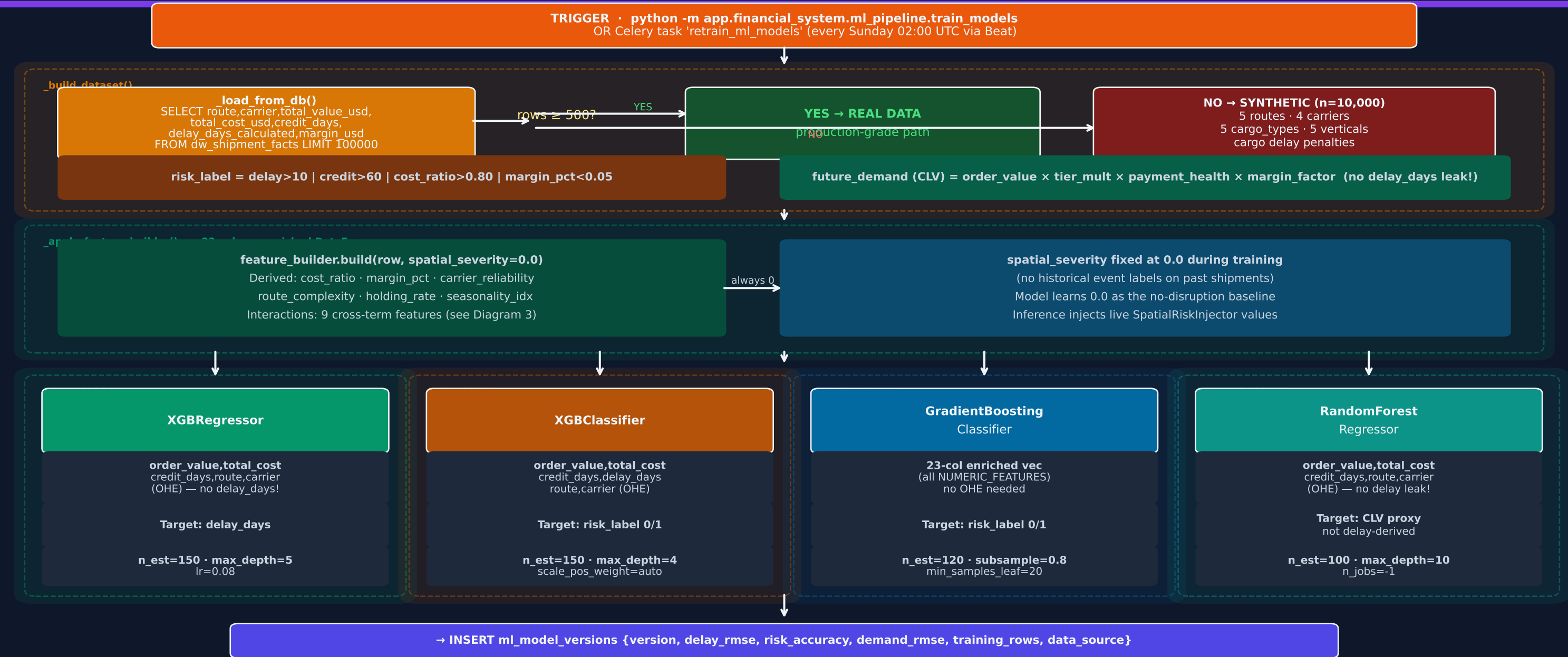
logit = cost_ratio×2.0 - margin_pct×4.0
+ delay_days×0.5 + risk_pressure×1.5

Always available · pure math
no file loads required

OUTPUT → risk_score: float [0,1] risk_label: HIGH / MEDIUM / LOW

FISCALOGIX

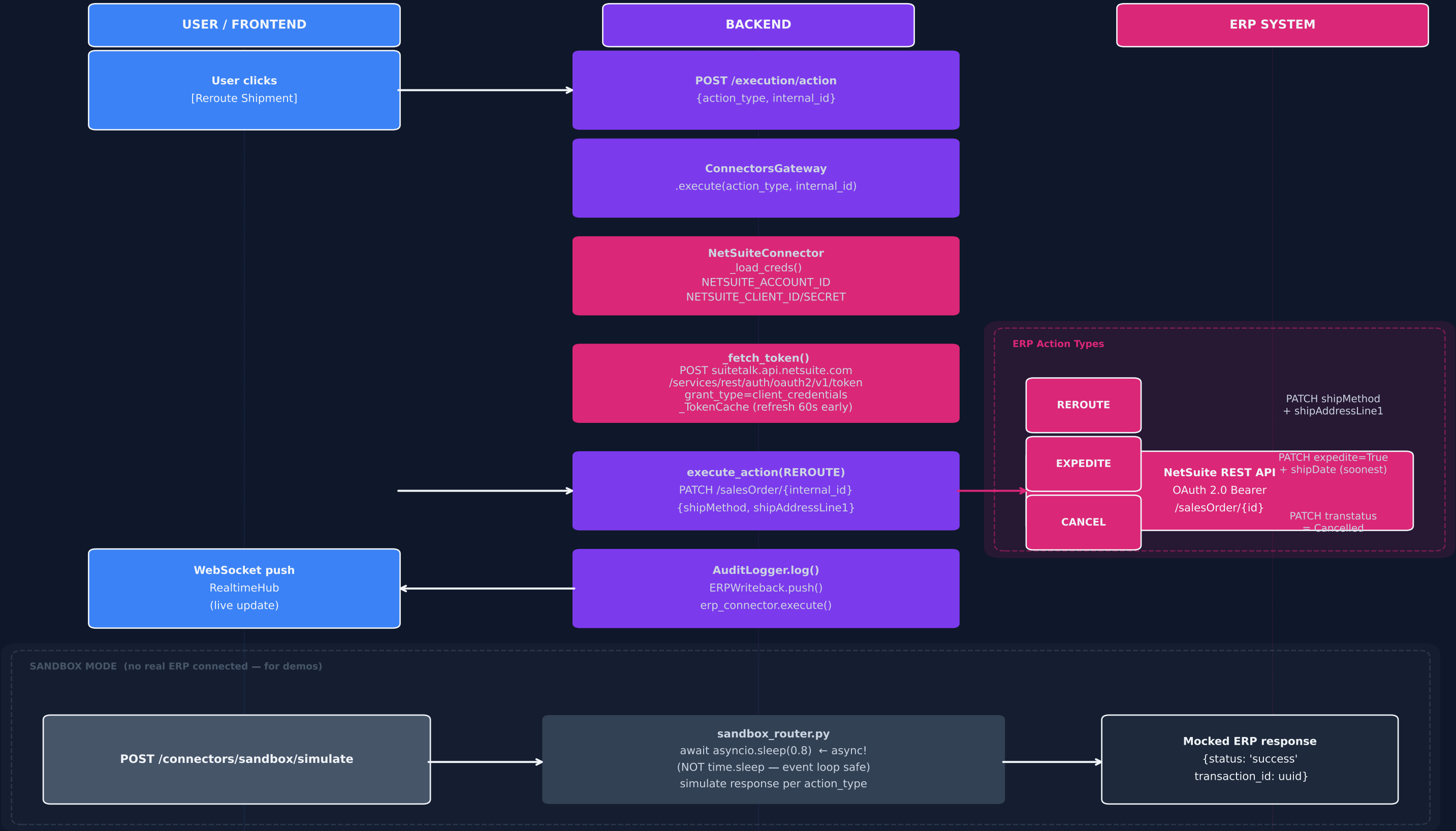
4 · ML Training Pipeline — Data Flow

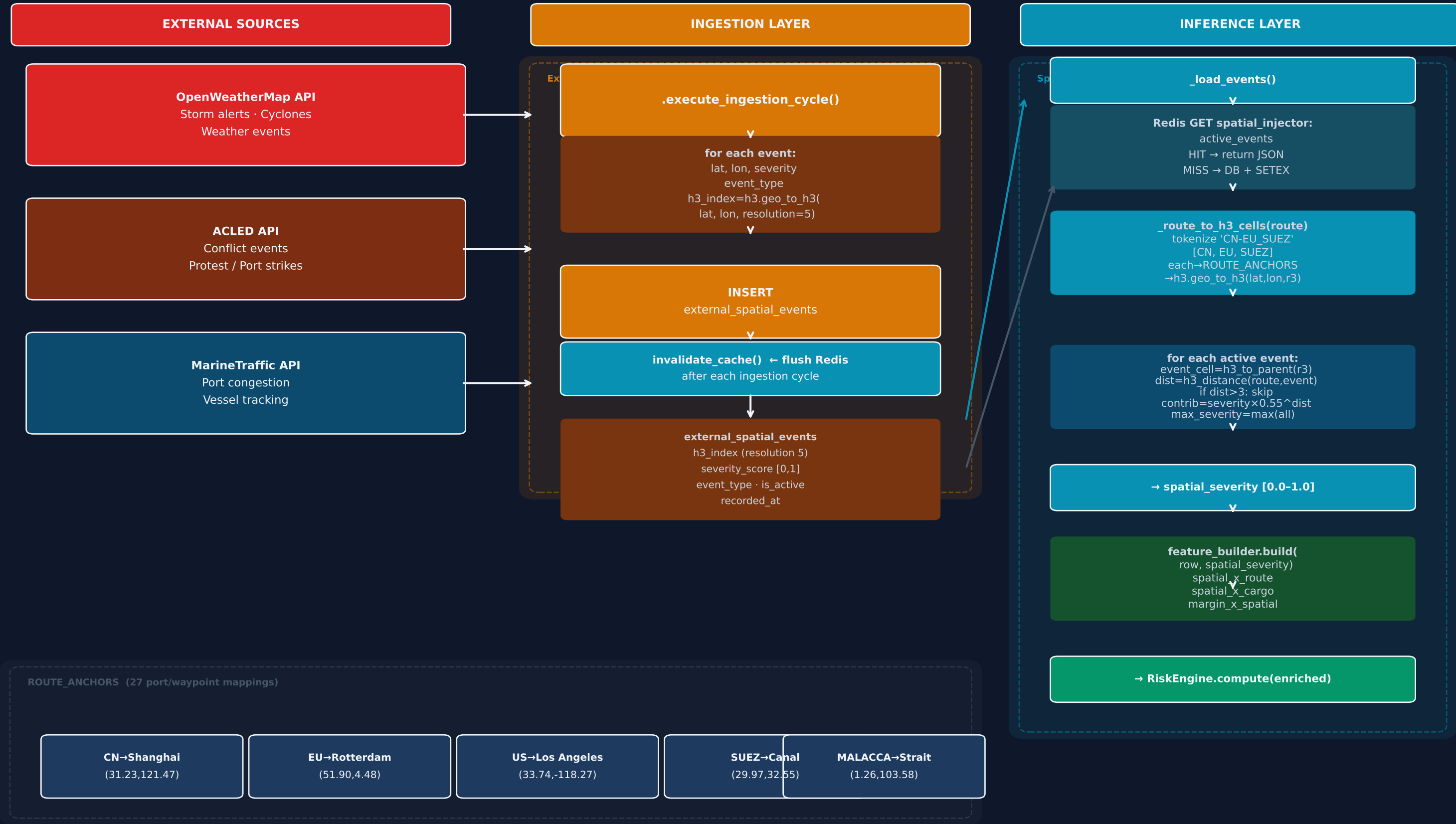


5 · Celery Async Task Topology

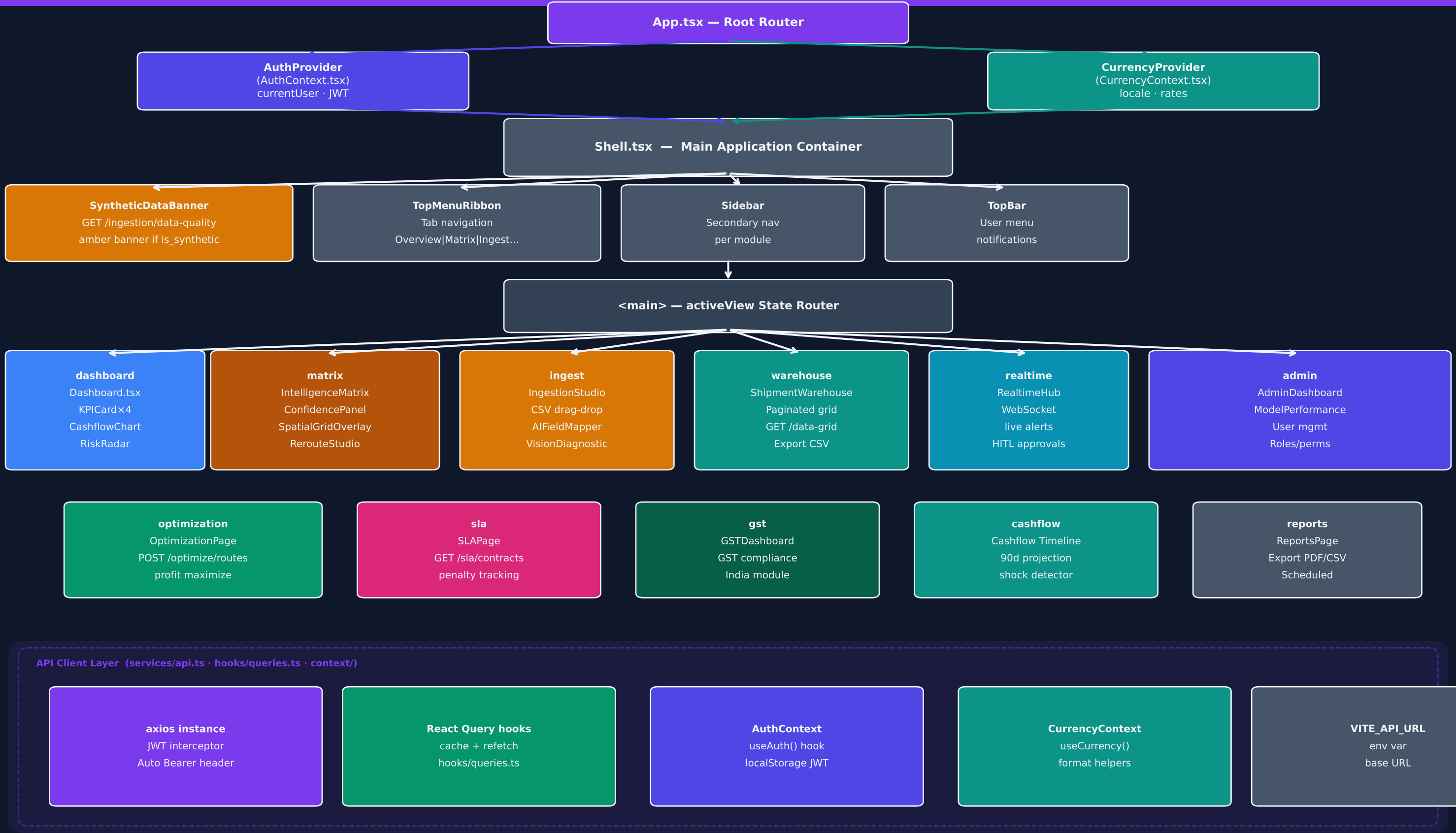


6 · ERP Connector Data Flow





8 · Frontend Component Architecture (React + Vite)



9 · Database Schema — Key Tables

PostgreSQL 15

dw_shipment_facts

id (PK)
tenant_id ← multi-tenancy key
source_system 'BULK-CSV' | 'ERP-SYNC'
route
carrier
total_value_usd
total_cost_usd
margin_usd
credit_days

delay_days_calculated

nlp_extracted_penalty_rate

recorded_at

external_spatial_events

id (PK)
h3_index (resolution 5)
severity_score NUMERIC [0,1]
event_type 'weather' | 'conflict' |
 'port_congestion'
is_active BOOLEAN
recorded_at

roles

id (PK)
name
permissions JSONB {
 can_approve: bool
 can_admin: bool
 can_write: bool
 can_view_costs: bool }

sla_contracts

id (PK)
tenant_id
contract_name
penalty_rate
penalty_type 'percentage' | 'flat_fee'
force_majeure_applies BOOLEAN
extracted_at

ml_model_versions

id (PK)
tenant_id
model_name
version 'v-2026-05-05'
is_active BOOLEAN
training_rows
data_source 'real' | 'synthetic'
delay_rmse
risk_accuracy
demand_rmse

users

id (PK)
email
hashed_password
tenant_id
role_id (FK → roles.id)

← FK

audit_log

id (PK)
user_id
tenant_id
action_type
payload JSONB
timestamp

Redis 7 (Cache + Broker)

fx_rates:{currency}

TTL: 55 min

Spot rates all currencies

open.er-api.com → ECB feed

wacc:{vertical}

TTL: 6 hours

WACC per industry vertical
adjusted vs FRED Treasury yield

spatial_injector:

active_events

TTL: 5 min

JSON list of active H3 events
flushed after each ingestion

celery broker

task queues

persistent

Beat schedule + API-triggered
task routing and results

session tokens

TTL: 24 hr

JWT token store

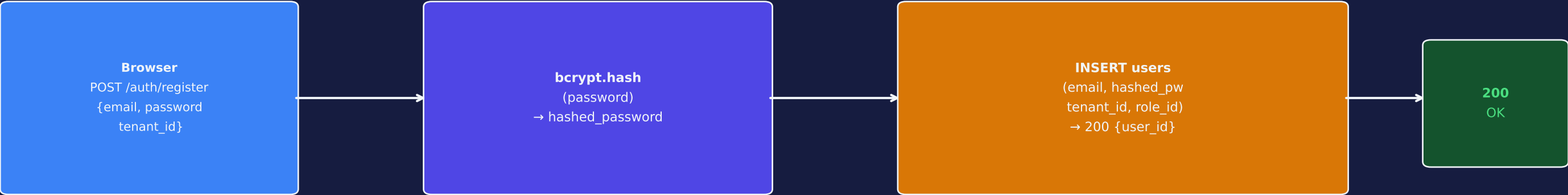
(logout blacklist support)

- Fact table
- Contracts
- Spatial events
- ML governance
- Auth/RBAC
- Users

10 · Deployment Architecture — Production EC2



REGISTRATION

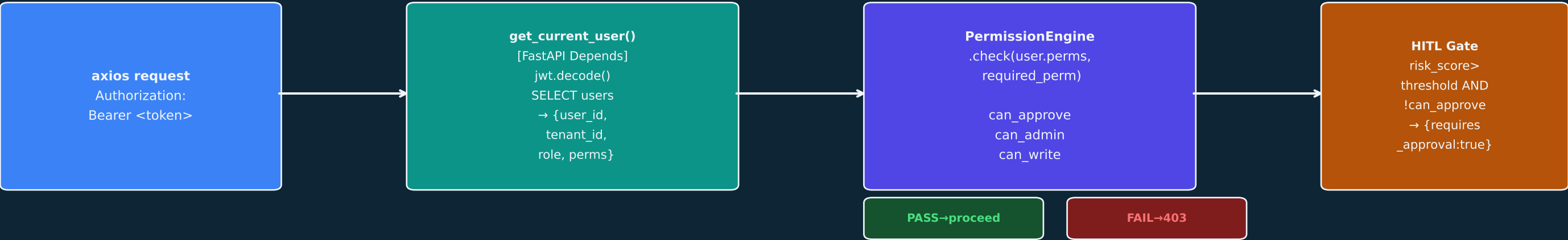


LOGIN



Frontend → localStorage.setItem('token', access_token) · axios interceptor → Authorization: Bearer {token}

EVERY PROTECTED REQUEST



12 · RAG (Retrieval-Augmented Generation) Pipeline

